

Comparison of Functional Outcome Measures After ACL Reconstruction in Competitive Soccer Players

A Randomized Trial

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Background: The choice of graft for anterior cruciate ligament (ACL) reconstruction remains controversial. Although many outcome studies comparing bone-patellar tendon-bone and semitendinosus and gracilis tendon grafts have been performed, most have not included tests of functional outcomes that challenge the graft. The objective of the current study was to compare the functional performances of soccer players after ACL reconstruction that was performed with either a bone-patellar tendon-bone or a semitendinosus and gracilis tendon graft.

Methods: Forty-two soccer players with unilateral ACL injury were prospectively randomized to a bone-patellar tendon-bone group (BPTB group; twenty-one subjects) or a semitendinosus and gracilis tendon group (STG group; twenty-one subjects) and followed a rehabilitation protocol. At the time of return to sports, the patients were asked to execute strength, hop, and jump tests and the results were compared between the groups. Twenty-one healthy athletes were recruited as the control group to evaluate their performance of the same tests.

Results: The limbs with the ACL reconstruction in the STG group had greater values for quadriceps torque as well as on the triple-hop, crossover-hop, and jump-landing test ($p < 0.01$), but the STG and BPTB groups showed similar results in terms of hamstrings peak torque and the results of two other hop tests ($p > 0.05$). Moreover, the subjects with ACL reconstruction had significantly lower values with regard to quadriceps and hamstrings peak torques and the results of the hop and jump tests compared with the healthy athletes ($p < 0.01$).

Conclusions: At the time of return to sports, the STG group had better performance in terms of quadriceps strength and the results of the triple-hop, crossover-hop, and jump-landing tests compared with the BPTB group. Compared with controls, soccer players who had undergone ACL reconstruction had less quadriceps and hamstrings strength and inferior hop performance and jump-landing strategy.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

The anterior cruciate ligament (ACL) is the most frequently injured knee ligament, accounting for about 50% of all knee ligament injuries¹. The majority of injuries occur during sports activities. Soccer is one of the sports with the highest prevalence of ACL injury^{2,3}. Athletes with an ACL injury who wish to participate in competitive sports are usually advised to undergo ACL reconstructive surgery⁴, which is typically performed with either a bone-patellar tendon-bone (BPTB) or semitendinosus and gracilis tendon (STG) graft.

The majority of athletes who sustain ACL injury do not successfully return to their preinjury sports level⁵⁻⁷. One major reason for this could be that the athletes do not regain their preinjury function⁸. Moreover, restored lower-extremity function is an important factor to minimize the risk of reinjury⁸. One of the important issues in the field of rehabilitation after ACL reconstruction is selecting the appropriate rehabilitative protocol and then deciding when to allow an athlete to return to sports. Subjective and functional outcomes according to the

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International Classification of Functioning, Disability and Health (ICF) have been suggested as appropriate criteria with which to evaluate the effectiveness of rehabilitation for athletes after ACL injury^{9,10}. The authors of previous studies comparing the results of BPTB reconstructions with those of STG reconstructions often have used outcomes such as knee range of motion¹¹, pain¹², strength¹¹, joint laxity¹¹, subjective questionnaires^{12,13}, and radiographic assessments¹¹. These commonly used outcomes may not be demanding enough to identify the possible differences between the function of athletes following the two surgical methods of ACL reconstruction and that of uninjured healthy subjects⁸.

To obtain a more complete assessment of function following ACL reconstruction, it has been suggested that batteries of functional tests that provide an indication of muscle strength and power, as well as neuromuscular control, should be used^{9,14}. A variety of these tests to measure the effectiveness of surgical and rehabilitative interventions have been described, including hop tests for distance, timed hop tests, figure-of-eight running tests, and vertical jump tests⁹. Keays et al.¹¹ suggested that the functional performance on running, sidestep, and single and triple-hop tests was better after reconstruction with an STG graft than it was after reconstruction with a BPTB graft at the six-year postoperative assessment. In contrast, Maletis et al.¹⁵ and Sajovic et al.¹⁶ reported no difference in the results of the single-hop test between the two groups; however, the single-hop test has been hypothesized as not being demanding enough to detect performance limitations. In many instances, an athlete who shows an acceptable result on a single functional test does not demonstrate acceptable function when he/she is evaluated with more demanding tests^{9,14,17}. Although many investigators have compared the clinical and subjective outcomes of the two surgical methods for ACL reconstruction, we are not aware of any study in which functional outcomes were used to investigate the results of the two surgical methods at the time when the patient returned to sports.

As there was, to our knowledge, no study comparing functional results between athletes who had undergone ACL reconstruction with a BPTB graft and those treated with an STG graft, we conducted a study to compare, in a randomized fashion, several functional outcomes in two groups of soccer players who had undergone ACL reconstruction with one of the two grafts; the outcomes included muscle strength and the results of hop and jump-landing tests and were performed at the time of return to sports (eight months postoperatively). Our hypothesis was that different functional outcomes would be obtained depending on whether a BPTB or STG graft had been used for the reconstruction. The second purpose of the study was to compare the performance of athletes who had undergone ACL reconstruction with those of matched controls.

Materials and Methods

Subjects

The subjects for this controlled study were selected from a potential cohort of sixty-one consecutive male athletes who met the inclusion criteria and were scheduled to undergo ACL reconstruction from 2010 to 2011. The athletes

were included if they had a unilateral ACL-deficient knee confirmed by physical examination and magnetic resonance imaging (MRI) and planned to return to sports. The exclusion criteria for subjects who had undergone ACL reconstruction were any other orthopaedic spine or lower-limb injuries (except meniscal injuries). Seventeen patients did not want to participate in the study, leaving forty-four patients. At the time of surgery, the patients were randomized, with use of a computerized random number generator, to receive either a BPTB graft or an STG graft. Two patients had a grade-IV chondral lesion at the time of surgery and were excluded, leaving forty-two patients randomized to the BPTB and STG groups (twenty-one patients in each). The sample size was estimated for the groups on the basis of power calculations for the primary outcome measures (hop and landing) from a pilot study. Considering a possible imprecision of the variability estimate, analysis showed that enrolling seventeen subjects in each group would provide an adequate sample size to achieve a power of 80% at the 5% significance level to detect a 5% difference (the difference believed to be clinically relevant).

All surgical procedures were performed by the same surgeon using an arthroscopic single-incision technique. After the ACL reconstructions, all subjects followed an accelerated rehabilitation protocol in a unique rehabilitation center (incorporating strengthening, balance, proprioception, stability, agility, plyometrics, and return-to-sport training) and intended to return to sports, which they did at a mean of eight months (range, six to nine months) postoperatively (Fig. 1). The time to return to sports was based on the demonstrated ability to do sport-related movements safely, determined by the surgeon and physical therapist. Participants in the ACL-reconstruction group were asked to bring one or two teammates with them to serve as the control group (comprising twenty-one individuals) matched for activity level and age. These healthy subjects had no recent history of lower-limb injury or surgery. Demographic characteristics of the groups are shown in Table I. All of the participants had a Tegner score of 9 (participation in competitive sports)¹⁸.

Details of the study were explained to the subjects, and informed consent was obtained. The study was approved by the Ethics Committee for Health Sciences Research, and the trial was registered in ClinicalTrials.gov (registry number NCT01693757).

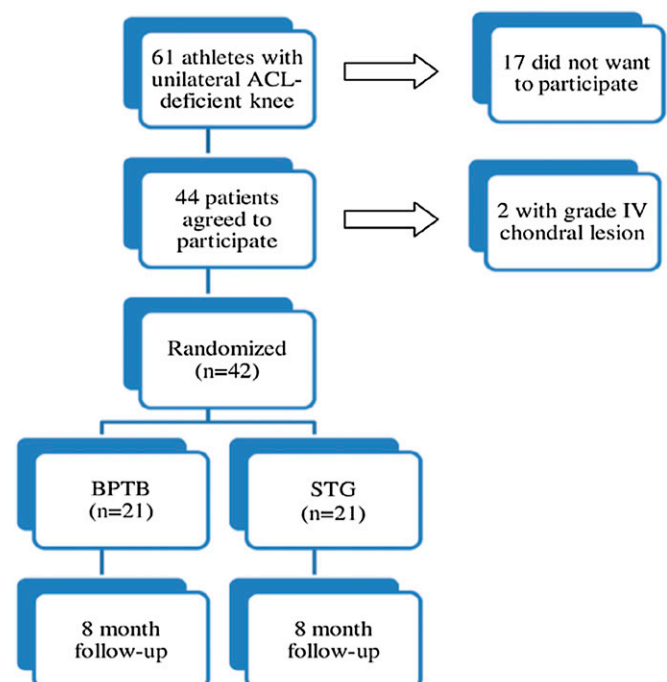


Fig. 1

Study participant flowchart based on the CONSORT diagram.

TABLE I Demographic Characteristics of the Three Groups

	BPTB (N = 21)	STG (N = 21)	Controls (N = 21)	P Value
Age (yr)				0.17
Mean (standard deviation)	24.9 (2.8)	25.2 (2.4)	25.4 (2.9)	
Range (median)	22-26 (24.5)	23-27 (25)	23-27 (25)	
Height* (cm)	178.3 (7.8)	176.9 (7.5)	179.7 (8.1)	0.14
Weight* (kg)	73.3 (8.5)	73.9 (7.6)	74.6 (7.5)	0.48
Time after surgery* (mo)	8.1 (1.4)	8.2 (1.8)	Not applic.	0.32
Operatively treated limb (dominant/nondominant)†	4/17	5/16	Not applic.	Not applic.
Arthroscopic partial meniscectomy‡	14	13	Not applic.	Not applic.
Mean Tegner score‡ (points)	9	9	9	Not applic.

*The values are given as the mean (standard deviation). †The values are given as the number of limbs. ‡The range of Tegner scores is from 0 to 10 points¹⁸.

Testing Protocol

Each subject was asked to perform three trials of quadriceps and hamstrings strength and hop and jump tests of the two limbs, conducted by the same evaluator in a randomized manner, and average scores were used for analysis. There was a one-minute rest after every test. The total trial duration was forty minutes. The nondominant limb of the controls was matched to the limb with the ACL reconstruction of the study patients because the majority of the reconstructed limbs were nondominant¹⁹. Leg dominance was determined by asking each subject to kick a ball; an examiner identified the nondominant limb on the basis of which one was not used to kick the ball.

Strength Testing

Quadriceps and hamstrings isokinetic strength was assessed with a dynamometer (Cybex International, Medway, Massachusetts) with the subject sitting and the pelvis, thigh, and ankle stabilized with straps. After a five-minute warm-up, the subject was asked to perform maximal extension-flexion concentric efforts, at the random velocities of 60°/s and 180°/s, and isokinetic peak torques were collected. The uninjured extremity of the subjects with an ACL reconstruction was tested first²⁰.

Hop Testing

Subjects were asked to perform a series of three hop tests for distance (single, triple, and crossover hop) and a 6-m timed hop test, which have been reported to be reliable and valid tests for patients with an ACL reconstruction²¹. The testing course consisted of a 6 m × 15 cm marking placed on the floor.

For the single hop, the participants stood on one leg, hopped, and landed on the same limb and the distance was measured. The triple hop consisted of three consecutive hops, and the total distance was recorded. For the crossover hop, the subjects hopped forward three times while alternately crossing over a marking, and the total distance was recorded. The timed 6-m hop consisted of one-legged hops in series for 6 m, and the time was measured by a stopwatch. The limb symmetry index was calculated for each of the four hop tests such that the score for the limb with the ACL reconstruction was expressed as a percentage of the score for the uninjured limb²¹.

Jump Landing

The participants were asked to drop off a 40-cm platform on one limb onto a 40 × 60 cm Kistler force plate (Kistler Instrument, Amherst, New York), immediately jump as high as they could (takeoff phase), and then contact the force plate with the same limb again (landing phase). The participants were allowed to use their arms freely for balance purposes. The 40-cm height was selected on

the basis of the findings by Huston et al.²², Ortiz et al.²³, and McNitt-Gray²⁴, who observed differences in landing mechanics from heights starting at 40 cm. The platform was sampled at 1200 Hz²⁵. Peak vertical ground reaction force (PVGRF) normalized to body weight was determined, and loading rate (PVGRF normalized to body weight divided by time to reach PVGRF) was collected as well^{26,27}. Raw data were filtered with a fourth-order Butterworth, zero-phase filter with a cutoff frequency of 6 Hz.

Statistical Analysis

The Statistical Package for the Social Sciences (SPSS) for Windows (version 16.0; SPSS, Chicago, Illinois) was used for statistical analysis. Age, height, weight, and time after surgery were compared among the three groups with use of one-way analysis of variance (ANOVA). To examine differences in peak torque, hop distance and time, and landing and takeoff PVGRF and loading rate, separate 3 × 2 (group by limb) mixed-model ANOVA was used to determine the main effects and interactions of the two factors for each variable. Separate ANOVA was used to determine the differences in limb symmetry index. Post hoc testing (Tukey) was performed to identify which pairwise comparisons demonstrated a significant difference. The level of significance was set at $p < 0.05$.

Source of Funding

There was no external funding source for this study.

Results

No significant difference in age, height, or weight was found among the groups (Table I). Moreover, there was no significant difference between the groups with the ACL reconstructions with regard to the time period following the surgery ($p = 0.32$). Mean and standard deviation values for quadriceps and hamstrings peak torques, hop-test results, limb symmetry indices, and jump-landing measures for both limbs in all groups are displayed in Tables II, III, and IV.

There was significant group × limb interaction for quadriceps peak torque ($p = 0.002$ for 60°/s and $p = 0.003$ for 180°/s), hamstrings peak torque ($p = 0.002$ for 60°/s and $p = 0.002$ for 180°/s), limb symmetry index ($p = 0.003$ for single hop and $p = 0.002$ for triple, crossover, and 6-m hops), and

TABLE II Peak Isokinetic Quadriceps and Hamstrings Muscle Torques at Velocities of 60°/s and 180°/s for the Athletes with ACL Reconstruction and Matched Controls

Test Group	Peak Torque: Treated Limb* (N/m)	P Value: BPTB vs. STG (Treated Limbs)	Peak Torque: Untreated Limb (N/m)
Quadriceps (60°/s)			
Control	189 (21)		188 (21)
BPTB	160 (21)	0.004†	182 (20)
STG	180 (19)		185 (22)
Quadriceps (180°/s)			
Control	138 (16)		137 (15)
BPTB	114 (12)	0.004†	134 (15)
STG	133 (13)		136 (14)
Hamstrings (60°/s)			
Control	103 (13)		102 (12)
BPTB	96 (13)	0.69	98 (17)
STG	97 (14)		100 (15)
Hamstrings (180°/s)			
Control	86 (11)		85 (11)
BPTB	81 (9)	0.63	82 (10)
STG	83 (11)		84 (11)

*The values are given as the mean (standard deviation). †A significant difference between the surgically treated limbs of the BPTB and STG groups.

landing and takeoff PVGRF and loading rate ($p = 0.002$). Further analysis revealed no significant differences between the surgically treated limbs of the two groups with an ACL

reconstruction with regard to hamstrings peak torque ($p = 0.69$ for 60°/s and $p = 0.63$ for 180°/s) or the limb symmetry index for the single-hop ($p = 0.78$) or 6-m-hop ($p = 0.74$) tests, but

TABLE III Results of Four Hop Tests on the Treated and Untreated Limbs of the Athletes with ACL Reconstruction and Matched Controls

Test Group	Treated Limb*	Untreated Limb*	Limb Symmetry Index*† (%)	P Value: BPTB vs. STG (Limb Symmetry Index)
Single hop (cm)				
Control	161.89 (10.4)	163.42 (11.3)	99.87 (6.4)	
BPTB	150.69 (12.4)	161.31 (10.7)	90.41 (7.9)	0.78
STG	151.24 (13.7)	161.63 (12.5)	90.57 (8.4)	
Triple hop (cm)				
Control	456.11 (34.3)	461.40 (35.1)	98.61 (8.5)	
BPTB	391.61 (32.4)	452.39 (37.5)	86.95 (8.5)	0.02‡
STG	419.54 (33.3)	458.28 (38.4)	90.49 (8.8)	
Crossover hop (cm)				
Control	424.14 (28.3)	426.32 (29.7)	97.91 (7.4)	
BPTB	363.68 (25.3)	421.24 (27.5)	85.52 (6.3)	0.02‡
STG	382.20 (32.1)	423.57 (30.8)	90.94 (8.1)	
6-m hop (s)				
Control	2.10 (0.4)	2.08 (0.4)	99.55 (6.4)	
BPTB	2.29 (0.8)	2.12 (0.6)	91.26 (7.3)	0.74
STG	2.28 (0.6)	2.10 (0.5)	91.39 (7.1)	

*The values are given as the mean (standard deviation). †Limb symmetry index = score for the limb with the ACL reconstruction/score for the healthy limb $\times 100$. ‡A significant difference between the limb symmetry index of the BPTB and STG groups.

TABLE IV Results of Jump-Landing Tests on the Treated and Untreated Limbs of the Athletes with ACL Reconstruction and Matched Controls

Test Group*	Treated Limb†	P Value: BPTB vs. STG (Treated Limbs)	Untreated Limb†
Landing PVGRF (N)			
Control	2.2 (0.4)	0.002‡	2.2 (0.3)
BPTB	1.5 (0.4)		2.5 (0.2)
STG	1.8 (0.5)		2.5 (0.2)
Takeoff PVGRF (N)			
Control	1.6 (0.3)	0.002‡	1.6 (0.1)
BPTB	1.0 (0.1)		1.5 (0.2)
STG	1.3 (0.2)		1.5 (0.1)
Loading rate (N-s)			
Control	24.1 (2.3)	0.002‡	23.6 (2.7)
BPTB	20.6 (3.3)		26.2 (2.5)
STG	22.5 (2.1)		26.5 (3.0)

*PVGRF = peak vertical ground reaction force. †The values are given as the mean (standard deviation). ‡A significant difference between the surgically treated limbs of the BPTB and STG groups.

the operatively treated limbs of the STG group had greater values for quadriceps peak torque (13% and 17% change, $p = 0.004$), triple-hop limb symmetry index (7% change) and crossover-hop limb symmetry index (5% change) ($p = 0.02$), and landing PVGRF (20% change) and takeoff PVGRF (30% change) and loading rate (9% change) ($p = 0.002$) compared with the BPTB group. There was no difference between the untreated limbs of the athletes with ACL reconstruction and the matched controls for any of the measures ($p > 0.05$) except landing PVGRF and loading rate ($p = 0.02$), which were higher for the athletes with ACL reconstruction. Overall, the operatively treated limbs of both groups had lower peak torques ($p = 0.01$), limb symmetry indices ($p = 0.002$), and PVGRF and loading rate ($p = 0.01$) compared with the matched controls.

Discussion

The most important finding of the present study was that there were differences in quadriceps strength as well as the results of the triple-hop, crossover-hop, and jump-landing tests, at the time of return to sports, between subjects who underwent ACL reconstruction with a BPTB graft and those treated with an STG graft. In addition, there was a significant difference in functional measures between the groups treated with ACL reconstruction and the matched controls.

The importance of using standardized outcomes for assessing patients treated with ACL reconstruction has been described repeatedly. It has been suggested that subjective and performance-based measures are necessary to quantify different aspects of function (or disability) of patients following ACL reconstruction and at the time of return to sports^{28,29}. Authors of previous studies comparing the results of reconstructions with a BPTB graft with those done with an STG graft often have used clinical and patient-oriented outcomes¹¹⁻¹³. To our

knowledge, the present study is the first to compare the functional outcomes of athletes at the time of return to the sports with a battery of functional tests to obtain a complete picture of the athletes' performance.

The current study showed no significant differences between the operatively treated limbs of the two groups with regard to hamstrings peak torque; however, the treated limbs in the STG group had greater values for quadriceps peak torque at the studied velocities compared with those in the BPTB group. Moreover, the treated limbs in the BPTB group had lower quadriceps peak torques compared with the matched limbs of the controls. The results are consistent with those in the study by Lautamies et al.²⁰, who found higher isokinetic quadriceps peak torque at the same velocities in patients treated with an STG graft, and Mohtadi et al.³⁰, who reported a trend toward loss of knee extension strength after reconstruction with a BPTB graft. Quadriceps muscle strength correlates with good outcome following ACL reconstruction³¹, and subjects with good quadriceps torque are able to return to their previous activity earlier³².

We also employed hop testing, a practical performance-based outcome measure that reflects the integrated effect of neuromuscular control, strength, and confidence in the limb and requires minimal equipment and time to administer³³. Reinke et al.³⁴ reported that the results of the triple-hop test were moderately correlated with patient-oriented outcomes such as the International Knee Documentation Committee (IKDC) and Knee injury and Osteoarthritis Outcome Score (KOOS), which are related to activity level. Fitzgerald et al.²⁸ suggested that the hop test might be a predictive tool for identifying patients who may have future problems as a result of injury. A limb symmetry index of $\geq 90\%$ ³⁵ has been classified as normal. The limb symmetry indices of the athletes with an ACL reconstruction in the current study were $>90\%$, except

for those in the triple-hop and crossover-hop tests in the BPTB group, which showed a noticeable difference compared with the STG and control groups. It has been reported that a single-hop test might not be demanding enough to detect performance limitations^{17,36}. Reid et al.²¹ mentioned that a combination of four hop tests might be a reliable and valid performance-based outcome measure after ACL reconstruction, since the tests incorporate a variety of movement principles that simulate the demands of dynamic knee stability. Our study showed that there were significant differences in triple-hop and crossover-hop results between the two types of surgery, with athletes who underwent reconstruction with an STG graft having higher values than those treated with a BPTB graft. These findings are comparable with the results reported by Keays et al.¹¹, who suggested that the functional results—triple hop and sidestep—of treatment with an STG graft were better than those of treatment with a BPTB graft; however, they evaluated their patients six years postoperatively. The reduced function following use of a BPTB graft may reflect the lower quadriceps strength and possibly the decreased ability of the quadriceps to perform plyometrically between hops¹¹. Itoh et al.³⁷ suggested that hop tests such as crossover and figure-of-eight are more demanding outcomes with which to evaluate and compare functional performances of patients following ACL reconstruction. Our results support our hypothesis that these functional tests might have the ability to differentiate between athletes treated with a BPTB graft and those treated with an STG graft.

Our study showed no difference in the results of the single-hop or 6-m timed hop test between the BPTB and STG groups. Maletis et al.¹⁵ and Sajovic et al.¹⁶ reported similar findings, but Keays et al.¹¹ found greater values for the single-hop test in subjects treated with an STG graft. Although single-hop tests have a weak correlation with patient-oriented outcomes and activity level³⁴ and a relatively low sensitivity for detecting functional differences between groups of patients with an ACL reconstruction³⁷, factors such as different samples and different periods after ACL reconstruction could explain the inconsistency in results.

Single-limb jump, a maneuver commonly used in sports such as soccer^{23,24}, is a functional test of neuromuscular performance that requires large eccentric quadriceps forces during the control of the joint flexion (deceleration) phase^{26,27}. Our results showed that the uninvolved limbs of the groups with an ACL reconstruction had greater landing PVGRF and loading rate than the involved limbs in those groups and the two limbs of the controls. Paterno et al.²⁶ found significant differences in dynamic-phase duration and peak torque between their ACL reconstruction and control groups. The combination of increased PVGRF and loading rate may place the untreated limb at greater risk for ACL injury following a return to sports³⁸. Moreover, our results showed higher jump-landing values in the treated limbs of the STG group compared with those in the BPTB group. To our knowledge, this difference in functional outcomes between the two types of surgery has not been described before as no study has compared the jump-landing patterns of athletes following ACL reconstruction. The reduced

function in our BPTB group may reflect the lower quadriceps strength that was shown in our study and previous studies¹¹.

The hypothesis of this study was partly supported by its results. Although there were no significant differences in the single-hop and 6-m-hop outcomes between the two graft groups, the STG group had better function in the more demanding functional tests at the time of return to sports. In addition, there were significant differences in functional measures between the athletes who had undergone ACL reconstruction and the matched controls. It appears that the dynamic function of the treated limbs of competitive athletes was not fully restored eight months after ACL reconstruction. This means that rehabilitation protocols should be further developed to be more effective and focus on more precise, demanding criteria for lower-limb function. Using batteries of functional tests may contribute to a more detailed evaluation of the athlete's function, which may aid in the decision regarding whether athletes can return safely to strenuous physical activities.

The current study had some limitations. First, it was specific to the point in time at which the athletes were ready to return to sports (eight months postoperatively) and the results are only valid for this period. Although short-term evaluation is critical for assessment with regard to return to sports, an assessment two to five years after surgery is essential to determine the long-term effect of surgery on maintaining functional outcome measures in competitive athletes. Moreover, this study included subjects with and without arthroscopic partial meniscectomy, which might affect the results.

In conclusion, at the time that the athletes returned to sports, the results of ACL reconstruction with an STG graft were better than those with a BPTB graft in terms of quadriceps strength and the results of triple-hop, crossover-hop, and jump-landing tests. Moreover, both groups with an ACL reconstruction demonstrated dynamic differences compared with matched controls. ■

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